

## Configuration System

The pipeline loads configuration from three layers, merged in order:

1. Defaults (config/defaults/) — version-controlled, baseline values for all settings
2. Local (config/local/) — gitignored, user-specific including credentials
3. Batch (config/batches/) — version-controlled, per-workflow overrides

## Setup

After cloning the repository:

1. Copy config/local/db\_config\_local.m.template to config/local/db\_config\_local.m
2. Edit db\_config\_local.m to add your database username and password
3. The file is gitignored and will not be committed

## Loading config in code

```
config = load_config();           % defaults + local, no batch overrides
config = load_config('migration'); % adds migration batch overrides
```

## Available batches

- migration — permissive thresholds for legacy data import (year\_min=1970, allows unknown lookup codes, points to migration\_overrides.csv)

## Adding a new batch

1. Create config/batches/<name>.m returning a struct with override fields
2. Create config/overrides/<name>\_overrides.csv for batch-specific warning overrides
3. Update this file to list the new batch

## Config struct sections

The merged config has three top-level sections:

| Section           | Description                                            |
|-------------------|--------------------------------------------------------|
| config.db         | Database connection settings (Type, Server, Port, ...) |
| config.validation | Validator thresholds and override behavior             |
| config.pipeline   | Pipeline settings (chunk_size, logging, known_fixes)   |

#### config.db

| Field        | Default       | Description                    |
|--------------|---------------|--------------------------------|
| Type         | 'MySQL '      | Database driver type           |
| Server       | 'localhost'   | Database host                  |
| Port         | 3306          | Port number                    |
| DatabaseName | 'NARWCDB '    | Target database                |
| DataSource   | 'NARWCDB_DSN' | ODBC DSN (SQL Server)          |
| Username     | ''            | Set in local/db_config_local.m |
| Password     | ''            | Set in local/db_config_local.m |

#### config.validation

| Field                              | Default | Notes                                                                      |
|------------------------------------|---------|----------------------------------------------------------------------------|
| datetime.year_min                  | 1970    | Records below this year are errors                                         |
| datetime.year_max                  | now+1   |                                                                            |
| datetime.year_warning              | 1980    | Records below this year trigger a warning                                  |
| coordinates.lat_min/max            | -90/90  |                                                                            |
| coordinates.study_area_lat_min/max | 20/55   | Warning if outside study area                                              |
| coordinates.study_area_lon_min/max | -85/40  |                                                                            |
| species.thresholds.group_size      | 100000  | Fallback when SPECCODE/TAXCODE have no threshold                           |
| species.thresholds.calf_count      | 100     | default                                                                    |
| environmental.visibility_allowed   | true    | Set to true in migration batch (legacy data has negative visibility codes) |
| warnings_become_errors             | false   |                                                                            |
| allow_unknown_lookup_codes         | false   | true in migration batch                                                    |
| overrides.enabled                  | true    |                                                                            |
| overrides.csv_path                 | ''      | Set in batch config                                                        |

#### config.pipeline

| Field                 | Default | Description                              |
|-----------------------|---------|------------------------------------------|
| chunk_size            | 10000   | Rows per processing chunk                |
| known_fixes.enabled   | true    | Apply apply_known_fixes.m pre-validation |
| logging.error_log_dir | 'logs/' | Directory for log files                  |

| Field                          | Default                         | Description                             |
|--------------------------------|---------------------------------|-----------------------------------------|
| logging.use_datetime_filenames | true                            | Stamp log filenames with run start time |
| logging.level                  | 'INFO'                          | Logging verbosity                       |
| format_definitions_path        | 'config/FormatDefinitions.json' | Path to format definitions              |
| lookup_tables_dir              | 'data/tables'                   | Lookup CSV directory                    |